

Foundation (Jalapeno)

Q1. Solve $|x| = 5$.

- (A) $x = 5$
- (B) $x = -5$
- (C) $x = 5$ or $x = -5$
- (D) $x = 0$
- (E) $x = 10$

Q2. Solve $|2x| = 8$.

- (A) $x = 2$ or $x = -2$
- (B) $x = 4$ or $x = -4$
- (C) $x = 1$ or $x = -1$
- (D) $x = 3$ or $x = -3$
- (E) $x = 0$

Q3. Solve $|x - 2| = 3$.

- (A) $x = 5$ or $x = -1$
- (B) $x = 2$ or $x = -2$
- (C) $x = 1$ or $x = -1$
- (D) $x = 4$ or $x = -4$
- (E) $x = 0$

Q4. Solve $|3x + 2| = 11$.

- (A) $x = 3$ or $x = -1$
- (B) $x = 2$ or $x = -2$
- (C) $x = 1$ or $x = -1$
- (D) $x = 3$ or $x = -5$
- (E) $x = 0$

Q5. Solve $|x + 1| = |2x - 3|$.

- (A) $x = 2$
- (B) $x = 1$
- (C) $x = 4$
- (D) $x = 2$ or $x = 1$
- (E) $x = 0$

Q6. Solve $|2x - 3| = |x + 2|$.

- (A) $x = 1$
- (B) $x = 2$
- (C) $x = 5$
- (D) $x = 1$ or $x = 5$
- (E) $x = 0$

Q7. Solve $|x| = |2x|$.

- (A) $x = 0$ or $x = 1$
- (B) $x = 0$ or $x = -1$
- (C) $x = 1$
- (D) $x = 0$
- (E) $x = 2$

Q8. Solve $||x| - 2| = 1$.

- (A) $x = 1$ or $x = -1$ or $x = 3$ or $x = -3$
- (B) $x = 2$ or $x = -2$
- (C) $x = 0$ or $x = 4$
- (D) $x = 1$ or $x = 3$
- (E) $x = 0$

Open Ended Questions (FRQ)

Q9. Solve $|x^2 - 4| = 5$ and check for extraneous solutions.

Q10. Solve $|2x - 3| = |x + 1|$ and check for extraneous solutions.

Chef's Tips

- Check for extraneous solutions.
- Use the definition of absolute value to solve equations.
- Graph the solutions to verify.